Hybrid Bell-State Optimizer

Concurrence Targeting with Final-State Fidelity

Anchor: Hybrid classical–quantum workflow for generating a Bell-like two-qubit state with specified concurrence.
Purpose: Demonstrate how a classical optimizer can steer a parameterized quantum circuit toward a desired entanglement target and then verify the final state by fidelity.

Boris Kiefer
New Mexico State University
bkiefer@nmsu.edu
GitHub
LinkedIn

Abstract. This brief describes a compact hybrid classical–quantum simulator for generating a two-qubit Bell-like state with a specified target concurrence. The quantum component is a parameterized circuit consisting of a single-qubit $R_y(\theta)$ rotation followed by a controlled-NOT gate. The classical component updates the rotation angle θ until the generated state's concurrence matches a user-defined target. After optimization, the simulator computes the fidelity between the optimized final state and the corresponding target state. Thus concurrence is the optimization objective, while final-state fidelity is the state-quality diagnostic. The simulator uses pure state vectors throughout; no density matrices are used.

Introduction

Hybrid classical–quantum algorithms combine parameterized quantum circuits with classical optimization. A quantum circuit prepares a state that depends on one or more adjustable parameters, and a classical optimizer updates those parameters according to a chosen objective function.

In this simulator, the objective is not an energy expectation value, but an entanglement measure: the concurrence of a two-qubit state. The goal is to generate a Bell-like state whose entanglement strength matches a prescribed target concurrence,

$$C_{\text{target}}.$$

After the optimizer finds a circuit parameter that produces the desired concurrence, the simulator evaluates the fidelity of the final state with respect to the intended target state. This gives two complementary outputs: concurrence verifies the amount of entanglement, while fidelity verifies state-level agreement with the target.

Important ordering. The simulator first optimizes concurrence. Fidelity is computed afterward as a final-state diagnostic. This mirrors the actual mathematical workflow in the code: generate state, compute concurrence, optimize θ, then compare the optimized state with the target state.

Theory: Parameterized Bell-Like State

The simulator begins with the two-qubit computational basis state

$$|00\rangle .$$

A single-qubit rotation $R_y(\theta)$ is applied to the first qubit:

$$R_y(\theta)\,|0\rangle = \cos\frac{\theta}{2}\,|0\rangle + \sin\frac{\theta}{2}\,|1\rangle .$$

After this rotation, the state is

$$|00\rangle \xrightarrow{R_y(\theta)} \left(\cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}|1\rangle\right)|0\rangle .$$

A controlled-NOT gate is then applied, using the first qubit as the control and the second qubit as the target:

$$\left(\cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}|1\rangle\right)|0\rangle \xrightarrow{\text{CNOT}} \cos\frac{\theta}{2}|00\rangle + \sin\frac{\theta}{2}|11\rangle .$$

Therefore, the generated state is

$$|\psi(\theta)\rangle = \cos\frac{\theta}{2}|00\rangle + \sin\frac{\theta}{2}|11\rangle .$$

This is a pure two-qubit state. The simulator does not use density matrices, mixed-state concurrence, noise channels, or partial traces.

Pure-State Concurrence

For a general pure two-qubit state

$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle ,$$

the concurrence is

$$C = 2|ad - bc|.$$

For the generated state,

$$|\psi(\theta)\rangle = \cos\frac{\theta}{2}|00\rangle + \sin\frac{\theta}{2}|11\rangle ,$$

the amplitudes are

$$a = \cos\frac{\theta}{2}, \qquad b = 0, \qquad c = 0, \qquad d = \sin\frac{\theta}{2}.$$

Substitution gives

$$C(\theta) = 2\left|\cos\frac{\theta}{2}\sin\frac{\theta}{2}\right| .$$

Using

$$2\sin\frac{\theta}{2}\cos\frac{\theta}{2} = \sin\theta,$$

the concurrence becomes

$$C(\theta) = |\sin\theta|.$$

Thus the rotation angle θ directly controls the amount of entanglement in the generated Bell-like state.

Target State and Final-State Fidelity

The user specifies a target concurrence,

$$0 \leq C_{\text{target}} \leq 1.$$

The corresponding target angle is

$$\theta_{\text{target}} = \arcsin(C_{\text{target}}),$$

using the principal solution

$$0 \leq \theta_{\text{target}} \leq \frac{\pi}{2}.$$

The corresponding target state is

$$|\psi_{\text{target}}\rangle = \cos\frac{\theta_{\text{target}}}{2}\,|00\rangle + \sin\frac{\theta_{\text{target}}}{2}\,|11\rangle\,.$$

After optimization, the simulator compares the optimized state

$$|\psi(\theta_{\text{opt}})\rangle = \cos\frac{\theta_{\text{opt}}}{2}\,|00\rangle + \sin\frac{\theta_{\text{opt}}}{2}\,|11\rangle$$

with the target state. The final-state fidelity is

$$F_{\text{final}} = \left|\langle\psi_{\text{target}}\rangle\,\psi(\theta_{\text{opt}})\right|^2.$$

For this specific circuit, the overlap is

$$\langle\psi_{\text{target}}\rangle\,\psi(\theta_{\text{opt}}) = \cos\frac{\theta_{\text{target}}}{2}\cos\frac{\theta_{\text{opt}}}{2} + \sin\frac{\theta_{\text{target}}}{2}\sin\frac{\theta_{\text{opt}}}{2}.$$

Therefore,

$$F_{\text{final}} = \left|\cos\frac{\theta_{\text{target}}}{2}\cos\frac{\theta_{\text{opt}}}{2} + \sin\frac{\theta_{\text{target}}}{2}\sin\frac{\theta_{\text{opt}}}{2}\right|^2.$$

Equivalently,

$$F_{\text{final}} = \cos^2\left(\frac{\theta_{\text{opt}} - \theta_{\text{target}}}{2}\right).$$

Success Metrics

The simulator reports two success metrics.

Concurrence Error

The concurrence error measures whether the generated state has the desired amount of entanglement:

$$\Delta C = \left| C(\theta_{\text{opt}}) - C_{\text{target}} \right| .$$

Since

$$C(\theta) = |\sin\theta|,$$

this becomes

$$\Delta C = \left| |\sin\theta_{\text{opt}}| - C_{\text{target}} \right| .$$

The concurrence target is successfully reached when

$$\Delta C < \epsilon_C .$$

Final-State Fidelity

The final-state fidelity measures whether the optimized state matches the intended target state:

$$F_{\text{final}} = \left| \langle \psi_{\text{target}} \rangle \, \psi(\theta_{\text{opt}}) \right|^2 .$$

The final state passes the fidelity check when

$$F_{\text{final}} > 1 - \epsilon_F .$$

A successful run therefore satisfies both

$$\Delta C < \epsilon_C$$

and

$$F_{\text{final}} > 1 - \epsilon_F .$$

In words, the simulator succeeds when the optimized circuit generates the correct amount of entanglement and the final state has high overlap with the target Bell-like state.

Hybrid Classical–Quantum Approach

The hybrid workflow consists of a quantum circuit evaluation followed by a classical parameter update.

The quantum part prepares

$$|\psi(\theta)\rangle = \cos\frac{\theta}{2}\,|00\rangle + \sin\frac{\theta}{2}\,|11\rangle .$$

The classical part computes the concurrence,

$$C(\theta) = |\sin\theta|,$$

and minimizes the loss

$$\mathcal{L}(\theta) = \left(C(\theta) - C_{\text{target}}\right)^2.$$

Equivalently,

$$\mathcal{L}(\theta) = \left(|\sin\theta| - C_{\text{target}}\right)^2.$$

The optimizer updates θ until the concurrence error falls below the selected tolerance. Once the optimized parameter θ_{opt} is obtained, the simulator computes the final-state fidelity,

$$F_{\text{final}} = \left|\langle\psi_{\text{target}}\rangle\, \psi(\theta_{\text{opt}})\right|^2.$$

Thus the complete logical flow is

$$\theta \longrightarrow |\psi(\theta)\rangle \longrightarrow C(\theta) \longrightarrow \mathcal{L}(\theta) \longrightarrow \theta_{\text{opt}} \longrightarrow F_{\text{final}}.$$

Pseudocode

Pseudocode: Hybrid Optimization with Final-State Fidelity

Choose target concurrence C_{target}
Choose concurrence tolerance ϵ_C
Choose fidelity tolerance ϵ_F
Define target angle:

$$\theta_{\text{target}} = \arcsin(C_{\text{target}})$$

Define target state:

$$|\psi_{\text{target}}\rangle = \cos\frac{\theta_{\text{target}}}{2}|00\rangle + \sin\frac{\theta_{\text{target}}}{2}|11\rangle$$

Initialize trial angle θ
while $|C(\theta) - C_{\text{target}}| \geq \epsilon_C$ **do**
    Prepare initial state $|00\rangle$
    Apply $R_y(\theta)$ to the first qubit
    Apply CNOT from the first qubit to the second qubit
    Obtain generated state:

$$|\psi(\theta)\rangle = \cos\frac{\theta}{2}|00\rangle + \sin\frac{\theta}{2}|11\rangle$$

    Compute concurrence:

$$C(\theta) = |\sin\theta|$$

    Compute loss:

$$\mathcal{L}(\theta) = (C(\theta) - C_{\text{target}})^2$$

    Update θ using the classical optimizer
end while
Store optimized angle θ_{opt}
Compute optimized state:

$$|\psi(\theta_{\text{opt}})\rangle = \cos\frac{\theta_{\text{opt}}}{2}|00\rangle + \sin\frac{\theta_{\text{opt}}}{2}|11\rangle$$

Compute optimized concurrence:

$$C_{\text{opt}} = |\sin\theta_{\text{opt}}|$$

Compute concurrence error:

$$\Delta C = |C_{\text{opt}} - C_{\text{target}}|$$

Compute final-state fidelity:

$$F_{\text{final}} = |\langle\psi_{\text{target}}\rangle\,\psi(\theta_{\text{opt}})|^2$$

Report θ_{opt}, C_{opt}, ΔC, $\mathcal{L}(\theta_{\text{opt}})$, and F_{final}

Interpretation

This simulator demonstrates how a classical feedback loop can steer a quantum circuit toward a desired entanglement property. The rotation angle θ controls the relative weights of the two basis states $|00\rangle$ and $|11\rangle$. Changing this angle changes the concurrence of the generated state.

When

$$\theta = 0,$$

the state is

$$|\psi(0)\rangle = |00\rangle\,,$$

and the concurrence is

$$C(0) = 0.$$

This is a separable product state.

When

$$\theta = \frac{\pi}{2},$$

the state is

$$|\psi(\pi/2)\rangle = \frac{1}{\sqrt{2}} \left(|00\rangle + |11\rangle\right),$$

and the concurrence is

$$C(\pi/2) = 1.$$

This is a maximally entangled Bell state.

Intermediate values of θ generate partially entangled Bell-like states. The optimizer searches for the value of θ that produces the requested target concurrence.

The final fidelity calculation checks a different property. Two states can have the same concurrence but still differ as state vectors. Therefore, after the concurrence objective is met, the simulator computes

$$F_{\text{final}} = \left|\langle\psi_{\text{target}}\rangle\, \psi(\theta_{\text{opt}})\right|^2$$

to verify that the optimized state is not merely equally entangled, but also aligned with the intended target state.

Takeaway

The simulator implements a mathematically transparent hybrid classical–quantum workflow. The quantum circuit prepares

$$|\psi(\theta)\rangle = \cos\frac{\theta}{2}\,|00\rangle + \sin\frac{\theta}{2}\,|11\rangle,$$

and the classical optimizer adjusts θ so that

$$C(\theta) = |\sin\theta|$$

matches the target concurrence.

The final output includes both the optimized concurrence and the fidelity of the optimized state with the target state:

$$F_{\text{final}} = \left|\langle\psi_{\text{target}}\rangle\, \psi(\theta_{\text{opt}})\right|^2.$$

Thus concurrence is the primary optimization target, while final-state fidelity is the state-quality check. Together, these quantities show whether the circuit generated the desired amount of entanglement and whether the resulting state matches the intended Bell-like target.